Stellar Cyber 6.6.0s is now live! Discover all the updates in the release notes!

Installing a Modular Sensor in Hyper-V

This topic describes how to install a Modular Sensor in a Hyper-V environment using the virtual disk image file downloaded from the Download Images tab in the System | DATA SOURCE MANAGEMENT | Sensors | Sensor Installation page. Refer to the following sections for details:

- **About Modular Sensors**
- **Site Preparation**
- **Creating A Virtual Switch**
- **Enabling Promiscuous Mode**
- **Obtaining the Virtual Disk Image**
- **Creating a Modular Sensor VM from the Disk Image**
- **Configuring the Sensor to Accept VLAN-Tagged Traffic**
- **Provision Processors**
- **Adding an Interface**
- **Applying a Token to the Installed Sensor**
- **Configuring a Static IP Address (Optional)**
- **Configure NTP and Set the Timezone**

Use our example as a guideline, as you might be using a different software version.

Stellar Cyber does not support the installation of third-party software on its virtual or physical device sensors.

About Modular Sensors

Sensors provide the data gathering foundation for Stellar Cyber's OpenXDR platform, gathering the right data with context. Modular sensors are purpose-built Stellar Cyber sensors that include both the host and the Stellar Cyber monitoring software. They are provided as both physical devices (Photon sensors) and virtual machine images for different target environments.

Previous releases provided a variety of different types of device sensors, including Network, Security, and Modular. Going forward, the only type of device sensor is Modular. You can use the Modular Sensor Profile to enable whatever sensor features you like, creating the same functionality provided by the different sensor types in previous releases.

A modular sensor lets you easily add the features you like to your sensor. This helps simplify your deployment and lets you manage the VM requirements for the sensors based on the modular features they use.

Modular Sensors always include log ingestion. From there, you can enable different features as part of your modular sensor profile:

- Enable the **Network Traffic** feature to monitor the virtual environment, the physical environment if connected to the span port of a physical switch, or the LAN segment via a mirror port on a switch. The sensor monitors network and server response times and can identify applications.

  The sensor converts that information to metadata and forwards it to the DP as Interflow. The DP can then provide security, DDoS, and breach attempt detections.
- Enable the **Sandbox** and **IDS** features to improve your security posture:
  - Sandbox lets you detect malware in files and network traffic through Stellar Cyber's integrated cloud service and also provides anti-virus services.
  - IDS lets you detect intrusion attempts using both files and network traffic.

Keep in mind that VM resource requirements increase as you add more features to the Modular Sensor Profile. Refer to **Virtual Modular Sensor Provisioning and Performance** for details on the resources required to run different combinations of features in a Modular Sensor Profile, as well as how to use the **show module** and **show module request** CLI commands to compare provisioned resources against those required to run specific feature combinations. Stellar Cyber only enables a Modular Sensor Profile on a sensor if the host VM's resources can support it.

Site Preparation

This process requires a Microsoft Hyper-V environment on a Windows server. Windows Server 2019 and 2016 are the tested versions.

Refer to **Virtual Modular Sensor Provisioning and Performance** for details on the resources required to run different combinations of features in a Modular Sensor Profile. Provision your modular sensor according to the features that you plan on enabling.

You will need:

-
-
- Open firewall ports for log ingestion.
-
- Open firewall ports for the features you plan on enabling in the Modular Sensor Profile for this sensor.

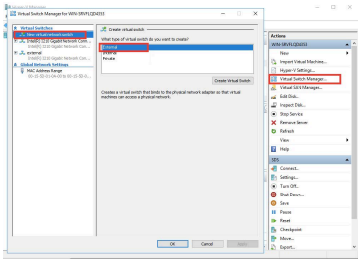
- Server switch with a physical network interface that supports promiscuous mode
- One IP address with access to a default gateway.
- A token obtained from the DP that will manage the sensor.

Creating A Virtual Switch

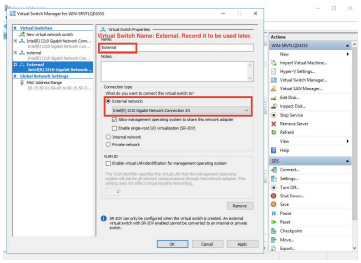
The first step is to create a virtual switch through which the Modular Sensor virtual machine communicates.

Use our example as a guideline, as you might be using a different software version.

1. Start the Hyper-V manager and open the Virtual Switch Manager from the right-hand sidebar. Select **Create Virtual Switch** then choose **External Network**. The resulting display is shown in the following image.



2. Select the physical network port that is to be used to connect to the outside world. The result appears similar to the sample in the following image.



Enabling Promiscuous Mode

You enable promiscuous mode so that the sensor can monitor all traffic. Promiscuous mode can't be enabled in the user interface, so the following commands must be performed in PowerShell.

In this example, the name of the switch is **External**. Change this value to the actual name of your switch in the commands.

```
C:\Users\Administrator> $a = Get-VMSystemSwitchExtensionPortFeature -FeatureId 776e0ba7-94a1-41c8-8f28-951f524251b5
C:\Users\Administrator> $a.SettingData.MonitorMode = 2
C:\Users\Administrator> add-VMSwitchExtensionPortFeature -ExternalPort -SwitchName <name of the switch> -VMSwitchExtensionFeature $a
```

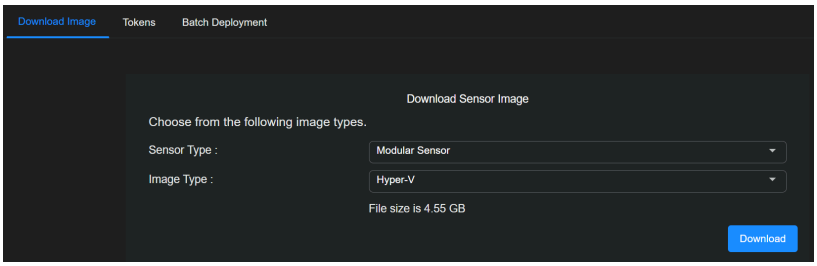
Obtaining the Virtual Disk Image

You download the virtual disk image for the Modular Sensor for Hyper-V from the **Download Images** tab in the **System | DATA SOURCE MANAGEMENT | Sensors | Sensor Installation page**. Use the following procedure:

Only users with the **System | DATA SOURCE MANAGEMENT | Sensors | Sensor Installation | Sensor Image Download** privilege assigned to their profile in the [the System | ORGANIZATION MANAGEMENT | Role-Based Access Privileges] interface can download images.

1. Navigate to the **System | DATA SOURCE MANAGEMENT | Sensors | Sensor Installation** page.
2. Set the **Sensor Type** dropdown to **Modular Sensor**.
3. Set the **Image Type** to**Hyper-V**.

   The display updates to show you the size of the file to be downloaded.

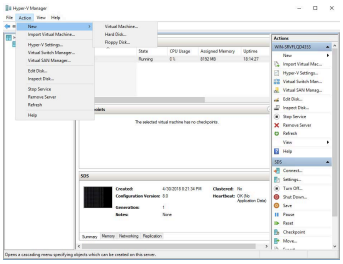


4. Click the **Download** button. The system downloads the virtual disk image (**aella-modular-ds-6.6.0.vhdx**) along with its corresponding SHA-1 hash file.

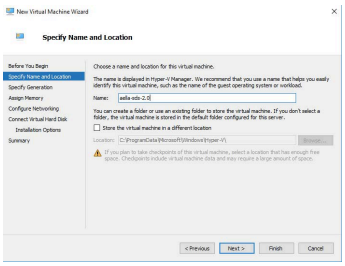
Creating a Modular Sensor VM from the Disk Image

Use the following procedure to create a Modular Sensor virtual machine from the virtual disk image you downloaded from the DP in the previous section.

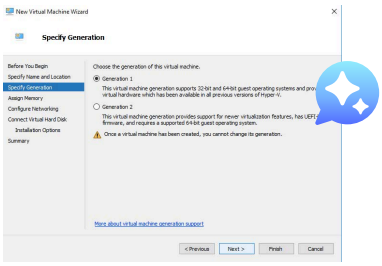
1. Return to the Hyper-V manager and select **Action** | **New** | **Virtual Machine...**.



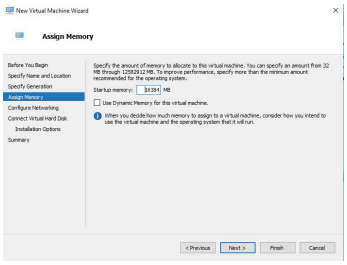
2. Specify a name for the new Modular Sensor. This can be any legal VM name but it is recommended to adopt a site convention so that the name identifies the type of sensor. The screen appears similarly to the image below.



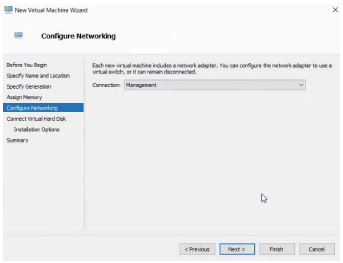
3. Enter the **Generation** parameter. The guest Operating System is a 64-bit OS, so **Generation 1** is a good choice, as illustrated below:



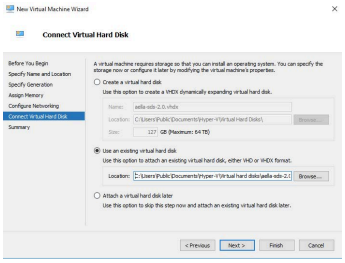
4. Next, assign memory according to the sensor's expected workload, as stated in **Server Sensor Sizing Specifications**.



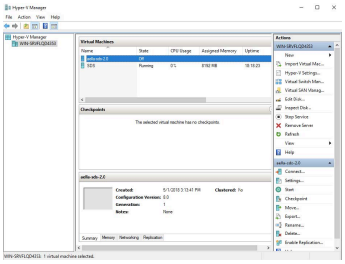
5. Configure the Management network interface. You can either enter a static IP or use a DHCP server. This is the interface that will be used for the sensor to send its Interflow data records to the data processor. The networking is configured as in the image below:



6. Connect the Virtual Disk image that you downloaded from the DP in **Obtaining the Virtual Disk Image**. Select the **Use existing** option, as illustrated below and use the Browse button to point the Location field to the downloaded **.vhdx** file.



7. The Modular Sensor VM appears in the Hyper-V Manager similarly to the image below:



Configuring the Sensor to Accept VLAN-Tagged Traffic

Capture interfaces for Modular Sensor VMs deployed in Hyper-V must be configured to operate in trunk mode when ingesting SPAN or mirror feeds that contain 802.1Q VLAN-tagged frames.

Trunk mode ensures that packets with 802.1Q VLAN tags from specified VLAN IDs are not dropped. This allows the sensor to analyze traffic from multiple VLANs simultaneously.

Configuration Procedure

1. Ensure that the SPAN or mirror feed from the switch arrives on the physical NIC of the Hyper-V host that is associated with an External Virtual Switch.
2. Configure the Modular Sensor VM capture interface to operate in trunk mode.

   Run the following PowerShell command on the Hyper-V host to configure the Modular Sensor VM capture interface in trunk mode with all VLAN IDs accepted:

   ```
   Set-VMNetworkAdapterVlan -VMName "<Sensor VM>" -Trunk -
   AllowedVlanIdList "1-4094" -NativeVlanId 0
   ```

   Parameter notes

   - **VMName** – Name of the sensor VM.
   - **Trunk** – Enables trunk mode on the VM network adapter.
   - **AllowedVlanIdList** – VLANs permitted to pass to the VM. In this example, we are accepting from the entire legal range of 802.1q VLAN IDs (1-4094). If only specific VLANs are required, replace `1-4094` with a comma-separated list or range. For example:

     `"10,20,30-40"`
   - **NativeVlanId** – Specifies the native VLAN (0 disables untagged assumptions).

Avoid Access Mode Configuration – Do not configure a single VLAN ID for the capture NIC. Setting a single VLAN ID places the adapter in access mode, which prevents other tagged VLAN traffic from reaching the VM.

Verification

After configuration, verify that VLAN tags are visible in captured traffic.

Example command:

```
tcpdump -i <interface> -e -nn vlan
```

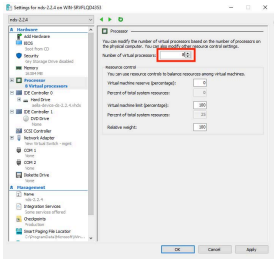
Expected result:

- Captured packets include 802.1Q VLAN tags.
- Traffic from the expected VLANs is visible.

Provision Processors

Before you power on the Modular Sensor VM, you must provision sufficient CPU cores to handle the Sensor's expected workload as summarized in in **Server Sensor Sizing Specifications**.

1. Select the VM in the list and select the **Settings** | **Processor** menu options. Then, change the number of processors and click **OK**.



Enabling SSSE3 and Disabling Processor Compatibility Mode

The sensor VM must have SSSE3 enabled and processor compatibility mode disabled in order for the Modular Sensor to operate correctly.

In most cases, SSSE3 will already be enabled. However, if you encounter issues with packet collection or Interflow data generation, you can use the instructions below to ensure that SSSE3 is enabled.

Click here to see instruction on how to enable SSSE3.

Sensors installed on Linux hosts must have SSSE3 enabled for their processors in order to operate correctly. This is true for Modular Sensors and Linux Server Sensors, as well as legacy Network and Security Sensors.
SSSE3 is typically supported/enabled for most vCPUs, but may not be for certain legacy AMD vCPUs. See below for instructions on enabling SSSE3.

To enable SSSE3 for a virtual machine:

1. Start the virtual shell (virsh)
2. Type the following command to edit the virtual machine's settings:

   `edit <virtual_machine_name>`
3. Locate the **<cpu>** section. It should appear similar to the following:

```
<cpu mode='custom' match='exact' check='partial'>
    <model fallback='allow'>SandyBridge</model>
......
 </cpu>
```

4. Add the following line to the **<cpu>** section:

   `<feature policy='require' name='ssse3'/>`

   When you are done, the **<cpu>** section should appear similar to the following:

```
<cpu mode='custom' match='exact' check='partial'>
    <model fallback='allow'>SandyBridge</model>
    <feature policy='require' name='ssse3'/>
............
  </cpu>
```

5. Save changes to the virtual machine and exist virsh.
6. Run the following command:

```
virsh define /etc/libvirt/qemu/<virtual_machine_name>.xml
```

Each virtual machine has a configuration .xml file. Typically, these files are stored under **/etc/libvirt/qemu**, but the location may be different for your system.

7. Stop and start the virtual machine in virsh.

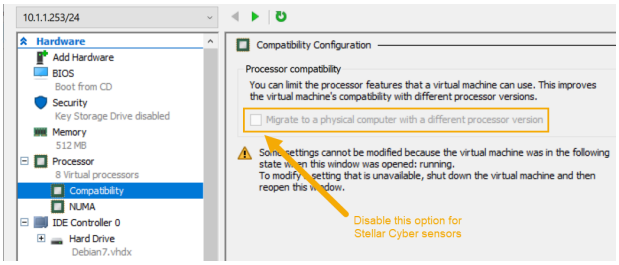
Ensure that Processor Compatibility Mode is Disabled

Stellar Cyber sensors deployed in Hyper-V must have the processor compatibility feature disabled to ensure that the SSSE3 instruction set can be used and that aella_flow runs properly. Although the sensor VM starts and appears to run correctly with processor compatibility enabled, stability issues will eventually occur due to high CPU usage.

You can verify that processor compatibility mode is disabled in the same dialog box where you just assigned CPU cores:

1. Expand the **Processor** entry in the left pane of the Settings dialog box.
2. Click on the **Compatibility** entry.
3. Ensure that **Migrate to a physical computer with a different processor** is disabled and **click** OK.

   The image below shows you the option that must be disabled for Stellar Cyber sensors:



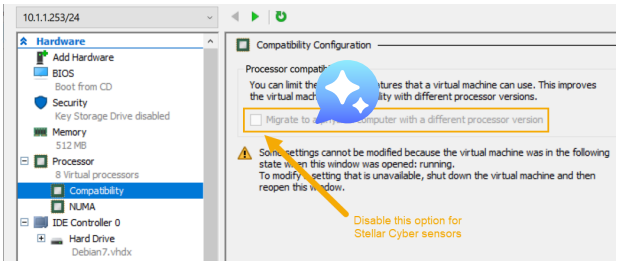
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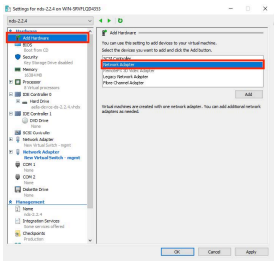


Adding an Interface

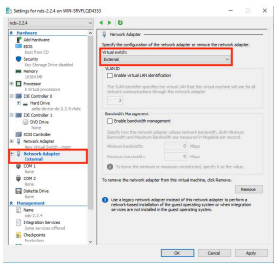
Before you power on the Modular Sensor VM, you must add a monitoring interface used to gather network data. You connect the monitoring interface to the virtual switch you added in [Creating A Virtual Switch]. Use the following procedure:

1. Click **Add Hardware** | **Network Adapter**.

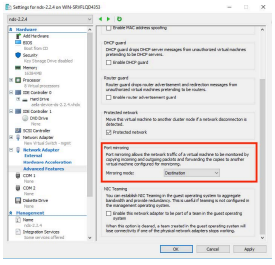
   You can only add a network adapter to the VM while it is powered off.



2. Select the virtual switch that you created in in [Creating A Virtual Switch], as illustrated below:



3. Expand the associated network adapter and click **Advanced Features**.
4. Set the mirroring mode of the aggregator to **Destination**, as illustrated below:



5. When you have finished adding the interface, power on the Modular Sensor VM.

Applying a Token to the Installed Sensor

The next step in installing a Modular Sensor is to obtain and apply the token used to authorize and configure the installed sensor.

Obtaining a Token for the Installation

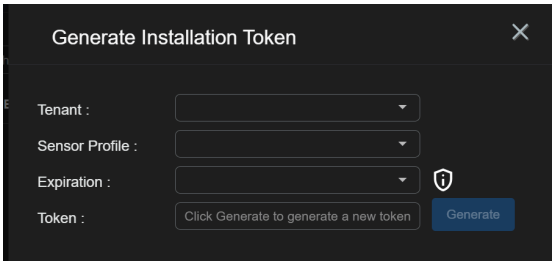
Tokens are required to authorize and configure the installation of a sensor image downloaded from the DP in the **System | DATA SOURCE MANAGEMENT | Sensors | Sensor Installation page**. Tokens point the installed sensor to the correct DP, assign the specified tenant, optionally provision a selected sensor profile, and authorize the sensor installation.

Use the following procedure to obtain a token in the Tokens tab:

1. Navigate to the **System | DATA SOURCE MANAGEMENT | Sensors | Sensor Installation** page and click on the **Tokens** tab.
2. If there is already an unexpired token that you want to use for the installation, you can either use the **Copy** button to copy it to the clipboard or use the **Download** button to download it as a file.
    - Copy the token if you plan on applying it by pasting it into a `set token string <token>` command in the CLI.
    - Download the token as a file if you plan on hosting the file on an HTTP server and referring to it in a `set token url <token url>` command.

    Refer to **Assigning Tokens** for a summary of the different ways in which tokens can be applied to a sensor installation.
3. You can also click the **Generate** button to create a new token.

    The Generate Installation Token dialog appears:

4. Select the tenant for the token from the **Tenant** dropdown. This is the tenant to which all sensors authorized with this token will be automatically assigned. The dropdown lists all tenants configured for your organization in the **System | ORGANIZATION MANAGEMENT | Tenants** page.
5. Select the Sensor Profile to be assigned to all sensors authorized with this token from the **Sensor Profile** dropdown. The dropdown lists all sensor profiles available in the System | DATA SOURCE MANAGEMENT | Sensors | Sensor Profiles page for the selected Tenant.

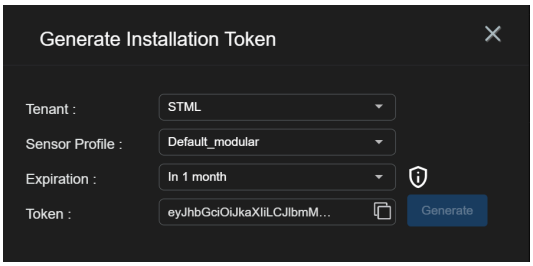
    If you do not want to assign a Sensor Profile with a token, you can set this field to **None (no sensor profile selected)**. This is also the setting for any tokens migrated from a pre-5.3.0 release as part of an upgrade.
6. Use the **Expiration** dropdown to select an expiration date for this token. You can select specified expiration dates ranging from two weeks to three months.

You can also select **Never expires** for the expiration date. However, Stellar Cyber recommends that you specify expiration dates for your tokens in order to improve the security of your deployment.

Remember that when a token expires, **sensors authorized with the token continue to operate as normal**. Once a sensor successfully registers with the Stellar Cyber platform, it no longer uses the token. It is only used for the initial authorization, registration, and configuration of the sensor.

7. Click the **Generate** button.

   The system generates the token and displays its contents in the **Token** field. The dialog also updates to display the expiration date for the token, as illustrated below.



8. You can use the **Copy** button to copy the token to the clipboard immediately, or simply close the dialog and retrieve the token from the **Tokens** tab later on.

Applying the Token to the Sensor

Tokens are required to complete the installation of a sensor image downloaded from the DP in the **Download Image** tab.

You apply tokens to sensors as the last step in the overall installation procedure:

1. Log in to your new Sensor. The default username/password is **aella**/**changeme**. You are immediately prompted to change the password.
2. Apply the token to the installed sensor from the sensor CLI with the `set token` command using **one** of the options in the table below:

   You only need to use **one** of the options in the table below. These are just two different ways to do the same thing – apply the token.

Option 1. Copy and Paste the Token String	Copy the token string from the **Tokens** tab and paste it into the CLI command. The syntax is as follows: `set token string <pasted string>`

Option 2. Host the Token on an HTTP Server	Download the token as a file from the **Tokens** tab, upload it to an HTTP server, and reference it in the `set token` command. The syntax is as follows:
	`set token url http://<url to token>`
	You can also use an HTTPS server. In that case, the specified URL must also include the username and password for the server using the following syntax:
	`set token url https://<user:password>@URL>`

3. The CLI reports that the **Sensor token is successfully set**.

   If you receive an e message instead, it's possible that the token has expired. Refer to the **Tokens** tab to see the expiration date. If you are using the File technique, it's also possible that an extra space or line may have crept into your text file – check the file to make sure it includes only the token text.

4. Wait a minute or so. Then, verify that the token was successfully applied using any combination of the following techniques:
   - Check the **System | DATA SOURCE MANAGEMENT | Sensors | Sensors** list in the user interface to see that the sensor has registered itself successfully.
   - Verify that the `show system` command shows all services as running.
   - Verify that the `show receiver` command displays a receiver.
   - Verify that the `show json` command reports some data sent in the `BYTE_SENT` column.

Configuring a Static IP Address (Optional)

By default, the sensor uses DHCP for the management port's IP address. For ease of troubleshooting, however, Stellar Cyber recommends that you reconfigure the management port to use a static IP address. The procedure is as follows:

1. Log in to your sensor. The default username/password is **aella**/**changeme**, but you changed this when you applied the token in the previous section.
2. You can set IP parameters manually using commands similar to the following (substitute your own IP parameters for the ones shown in bold below):

   ```
   set interface management ip 192.168.14.100/255.255.255.0
   set interface management gateway 192.168.14.1
   set interface management dns 8.8.8.8
   ```

3. Verify the IP settings with the `show interfaces` command.
4. Log out with the `quit` command.

Configure NTP and Set the Timezone

Stellar Cyber strongly recommends that configure NTP and set the timezone for the sensor.

Refer to **Best Practices for NTP and Timezones** for details.

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